Jülide Kahkeci

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EDUCATION

09/2021–09/2025

PhD in Environmental Engineering
University of Alberta | Edmonton, AB | GPA: 4.00/4.00
Thesis: *Biochar-Supported Bismuth Tungstate Photocatalysts for the Removal of Emerging Contaminants from Wastewater*
Relevant coursework: Physical/chemical water and wastewater treatment; Biological wastewater treatment; Advanced water and wastewater treatment processes; Municipal solid waste management

09/2018–08/2020

MSc in Materials Science and Nanotechnology
Bilkent University | Ankara, Turkey | GPA: 3.79/4.00
Thesis: *Microfluidic waveguides and solution lasers of colloidal semiconductor quantum wells*
Relevant coursework: Nanoscience and nanotechnology; Introduction to micro and nanofabrication; Computational methods for material science; Applications of microfluidics and nanofluidics

09/2014–06/2018

BSc in Chemical Engineering
Middle East Technical University | Ankara, Turkey | GPA: 3.72/4.00 (graduated first in rank)
Research: *Organic solvent nanofiltration membrane synthesis and application in wastewater treatment*
Relevant coursework: Heat and mass transfer; Thermodynamics (I, II); Membrane processes; Process control; Chemical reaction engineering; Separation processes; Mathematical modeling

RESEARCH EXPERIENCE

11/2025–Present

Postdoctoral Research Fellow, Water and Wastewater Treatment
University of Alberta | Edmonton, AB
- Prepare proposal, award, and grant applications to secure research funding
- Co-supervise 3 PhD students and mentor graduate students, providing guidance on experimental design, data interpretation, and manuscript writing and preparation
- Train lab members on instrument operation and laboratory procedures
- Collaborate with industry partners on 2 projects to design treatment systems, evaluate long-term pilot-scale performance, document findings, and propose recommendations for full-scale implementation
- **Research projects**:
  - Production of forestry-waste biochar for oil sands process water treatment in pilot-scale columns and constructed wetlands (funded by the Canada First Research Excellence Fund)
  - Development of solar and UV-driven photocatalysts and photocatalytic reactors for water treatment
  - Immobilization of photocatalysts for integration in flow-through reactor modules
  - Design of biochar and hydrochar adsorbents for metal adsorption

09/2021–09/2025

Graduate Research Assistant, Water and Wastewater Treatment
University of Alberta | Edmonton, AB
- Conducted literature reviews and identified research gaps in solar-driven photocatalytic technologies for municipal and industrial wastewater treatment
- Designed and synthesized biochar-supported photocatalysts, utilized advanced characterization techniques to correlate properties with performance, and assessed toxicity and leaching potential in real water matrices
- Developed and validated HPLC analytical methods, prepared standard operating procedures, delivered training sessions, and performed instrument calibration, maintenance, and troubleshooting
- **Achievements**:

  - Achieved 74-fold improvement in photocatalytic activity and demonstrated excellent removal of mixtures of emerging contaminants from river water, stormwater, and secondary effluent matrices
  - Authored a review article and 5 research articles in Q1 journals, and presented findings at the Canadian Society of Civil Engineering Conference in 2024

09/2020–08/2021

Graduate Research Assistant, Green Hydrogen Production

King Abdullah University of Science and Technology | Jeddah, Saudi Arabia

- Evaluated the integration of solar and wind energy for powering green hydrogen production from saline water using polymer electrolyte membrane (PEM) and alkaline electrolysis systems
- Developed system designs and mass and energy balances, and conducted techno-economic assessments to compare process configurations and energy-supply options
- Presented findings at the KAUST Research Conference: Near-Zero Carbon Combustion Technology (2021)

09/2018–08/2020

Graduate Research Assistant, Microfluidics

Bilkent University | Ankara, Turkey

- Designed and fabricated microfluidic devices for lasing applications, gaining expertise in nanofabrication, photolithography, thin-film deposition, and optical characterization
- Synthesized semiconductor colloidal quantum wells and integrated them in microfluidic lasers and waveguides
- Published results in Advanced Materials and presented findings at UNAM Nanoday 2019

09/2017–08/2018

Undergraduate Research Assistant, Chemical Engineering

Middle East Technical University | Ankara, Turkey

- Fabricated cellulose membranes via the alkaline hydrolysis of cellulose acetate for organic solvent nanofiltration and tested for industrial wastewater treatment applications
- Presented findings at Gazi University Chemical Engineering Conference 2018

TEACHING EXPERIENCE

01/2025–04/2025

Course Instructor, Environmental Engineering Design (CIV E 429)

University of Alberta | Edmonton, AB

Senior undergraduate-level course in drinking water and wastewater treatment facility design for indirect potable reuse, covering coagulation, filtration, membrane separation, biological treatment, and disinfection. Course includes field trips, design project technical reports, and presentations.

01/2023–12/2024

Graduate Teaching Assistant

University of Alberta | Edmonton, AB

- Supported the delivery of core environmental engineering courses in classroom and laboratory settings
- Delivered simulation software (BioWin, WaterGems) tutorials, held office hours, and provided one-on-one instruction to support student learning
- Evaluated assignments and technical reports, supervised laboratory work, and led site visits to water and wastewater treatment facilities
- **Courses**:
  - Biological Processes (ENV E 324): Fall 2024
  - Environmental Chemistry (ENV E 220): Fall 2024
  - Environmental Water Quality Management (CIV E 525): Fall 2024
  - Principles of Environmental Modelling and Risk (CIV E 321): Winter 2023, Winter 2024
  - Chemical and Physical Processes (ENV E 325): Winter 2023, Winter 2024
  - Fluid Mechanics (CIV E 330): Fall 2023
  - Municipal Systems (CIV E 421): Fall 2023

TEACHING DEVELOPMENT

04/2026

Graduate Teaching and Learning Program, University of Alberta, Level 3: Pedagogy and Course Design

01/2026

Graduate Teaching and Learning Program, University of Alberta, Level 1: Foundations

SERVICE

Editorial and Peer Review

2026–Present

Scientific Editor

Elsevier, Journal of Hazardous Materials Advances

2026–Present

Peer Reviewer

Processes, Applied Sciences, Water, Environment

Professional Service

2026–Present

Publishing Team Lead

2027 IWA Young Water Professionals Canada Conference

08/2026

Oral Presentation and Poster Judge

Faculty of Engineering Graduate Research Symposium

07/2026

Volunteer

5th International Conference on Disinfection & Disinfection Byproducts

OTHER EXPERIENCE

09/2022–04/2024

Exam Proctor, Accessibility & Accommodations Services

University of Alberta | Edmonton, AB

- Assisted in exam preparation and ensured that required materials (including USBs, supplementary resources, and specific instructions) were provided according to individual student accommodation requirement
- Communicated with students, proctors, and instructors to resolve exam-related concerns
- Verified student ID and exam details, coordinated exam filing and distribution, and maintained accurate records

09/2017–05/2018

Process Design (Capstone Projects)

Middle East Technical University | Ankara, Turkey

- Developed process flow diagrams and simulated reactor (Aspen HYSYS) and facility designs (Aspen Plus)
- Optimized processes and delivered technical reports with process flow diagrams, mass and energy balances, equipment sizing and selection, cost estimates, economic evaluation, and regulatory compliance analysis
- Projects:
  - Solvent-recovery facility to treat 100,000 m^3/h of solvent-laden air, recovering 400 kg/h of organic solvents and producing five high purity product streams
  - Styrene production facility (150,000 ton/y capacity) from methanol and toluene. Achieved >99.8% conversion and met design targets of 93% styrene purity through the recovery of unreacted feed in downstream separation to improve process efficiency

07/2016–09/2017

Process Engineering Internships
SCK Zeta Kimya (Personal Care Products facility) | Istanbul, Turkey
Rose Paints | Jeddah, Saudi Arabia
Eti Soda (Soda Ash facility) | Ankara, Turkey

TECHNICAL SKILLS

Modelling and Simulation: Aspen (Plus, HYSYS), BioWin, Solidworks, WaterGEMS
Programming and Data Analysis: MATLAB, R, Mathcad, Origin
Analytical Skills: High-performance liquid chromatography (HPLC), UV-Vis absorption spectroscopy, total organic carbon (TOC), chemical oxygen demand (COD), ion chromatography, physicochemical water-quality analyses (pH, conductivity, turbidity, alkalinity, dissolved oxygen, solids measurements)
Nanofabrication and Microfabrication Tools: Photolithography, soft lithography, physical vapor deposition (thermal evaporation, radiofrequency sputtering)
Material Characterization: X-ray diffraction, X-ray photoelectron spectroscopy, Fourier transform infrared spectroscopy, electron microscopy (SEM, TEM), surface area analysis (Autosorb), photoluminescence spectroscopy, electrochemical impedance spectroscopy, photocurrent response measurement, electron paramagnetic resonance spectroscopy, UV-visible diffuse reflectance spectroscopy

LANGUAGES

English, Arabic, Turkish

AWARDS AND RECOGNITION

07/2026 Best Oral Presentation Award, 5th International Conference on Disinfection & Disinfection Byproducts
06/2026 Graduate Student Association Academic Travel Grant ($500)
05/2026 Future Energy Systems: Graduate Student and Postdoctoral Scholar Opportunity Award ($2,500)
05/2026 Postdoctoral Fellows Association Travel Award ($500)
11/2025 Highly Qualified Professional, Canada First Research Excellence Fund ($60,000)
09/2024 Alberta Innovates Graduate Student Scholarship ($30,000)
09/2024 Alberta Graduate Excellence Scholarship ($12,000)
09/2024 Gordon R Finch Memorial Graduate Scholarship in Environmental Engineering ($2,700)
09/2024 Jacob H Masliyah Graduate Award in Oil Sands Engineering ($2,000)
09/2021 University of Alberta Graduate Recruitment Scholarship
09/2018 Bilkent UNAM Graduate Scholarship
06/2018 Middle East Technical University: First in rank certificate

PROFESSIONAL AFFILIATIONS

05/2026 Engineer-in-Training (E.I.T.), Association of Professional Engineers and Geoscientists of Alberta
05/2025 Young Water Professional, International Water Association Canada

PUBLICATIONS AND SCHOLARLY WORK

Peer-Reviewed Journal Articles

1. **Kahkeci, J.**, Sánchez-Montes, I., and Gamal El-Din, M. (2026). Tuning Pyrolysis Temperature of Wheat Straw Biochar Supports for Enhanced Bi_2WO_6 Photocatalytic Degradation of Stormwater Contaminants in Real Water Matrices. *Journal of Hazardous Materials*, 512, 142241.
2. **Kahkeci, J.**, Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (2025). Softwood Biochar-Supported Bi_2WO_6 for Photocatalytic Degradation of Organic Contaminants in River Water: Role of Pyrolysis Temperature and Surface Functionality. *Journal of Colloid and Interface Science*, 139132.

3. Aoudi, B., **Kahkeci, J.**, Sánchez-Montes, I., Boluk, Y., and Gamal El-Din, M. (2025). Efficient solar-driven degradation of a tire wear pollutant using floating K-doped g-C_3N_4 photocatalyst in secondary municipal wastewater. *Separation and Purification Technology*, 364, 132523.
4. Aoudi, B., **Kahkeci, J.**, Boluk, Y., and Gamal El-Din, M. (2024). Enhanced g-C_3N_4 for sustainable solar degradation of 1,3-diphenylguanidine (DPG) in wastewater: Investigating the effects of precursor, temperature, and potassium doping. *Chemical Engineering Journal*, 487, 150665.
5. **Kahkeci, J.**, Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (2024). Engineered biochar supported bismuth tungstate: Unveiling the influence of precursor concentrations and biochar dosage for the solar photocatalysis of 1,3-diphenylguanidine in secondary municipal effluent. *Chemical Engineering Journal*, 483, 149142.
6. **Kahkeci, J.** and Gamal El-Din, M. (2023). Biochar-supported photocatalysts: Performance optimization and applications in emerging contaminant removal from wastewater. *Chemical Engineering Journal*, 476, 146530.
7. **Maskoun, J.**, Gheshlaghi, N., Isik, F., Delikanli, S., Erdem, O., Erdem, E. Y., and Demir, H. V. (2021). Optical microfluidic waveguides and solution lasers of colloidal semiconductor quantum wells. *Advanced Materials*, 33(10), 2007131.

Manuscripts Submitted

1. Aoudi, B., **Kahkeci, J.**, and Gamal El-Din, M. (2026). Photocatalytic treatment of oil sands process water: recent advances, reactive pathways, treatment performance, and future opportunities. Submitted to *Current Opinion in Chemical Engineering*.
2. Mokarizadeh, H., Aoudi, B., **Kahkeci, J.**, Ganiyu, S., and Gamal El-Din, M. (2026). Solar-Driven Photocatalysis for Water and Wastewater Treatment: Material Trends, Mechanistic Insights, and Emerging Challenges. Submitted to *Environmental Science and Pollution Research*.

Manuscripts in Preparation

1. Gamal El-Din, M., Usman, M., Ganiyu, S., **Kahkeci, J.**, Saeed, S., and Bekele, A. (2026). Treatment of High Organic Laden Process Wastewater via Adsorption on Oily Sludge-Derived Chars Coupled with Electro-oxidation.
2. **Kahkeci, J.**, Aoudi, B., Li, Y., Stafford, J., and Gamal El-Din, M. (2026). Comparative Performance and Toxicity Assessment of g-C_3N_4, Bi_2WO_6, and ZnO Photocatalysts for Stormwater Treatment.
3. Fan, Y., **Kahkeci, J.**, Usman, M., Feng, Q., Li, Y., Bekele, A., Arslan, M., Jaffe, B., Jessup, R. W., Bremer, E., and Gamal El-Din, M. (2026). Influence of Torrefied and Pyrolyzed Biomass Amendments on Nutrient Dynamics and Soil Fertility in Clayey and Sandy Soils.
4. Fan, Y., **Kahkeci, J.**, and Gamal El-Din, M. Evaluating Forestry-Residue Biochars for Oil Sands Process-Affected Water Treatment in Constructed Wetlands: The Role of Adsorption and Microbial Degradation.
5. Al Mamun, M. R., **Kahkeci, J.**, Bunda-Fajunio, M. G., Gamal El-Din, Y., Hussain, N. A. S., Stafford, J. L., and Gamal El-Din, M. (2026). Solar Activation of Peroxymonosulfate for Micropollutant Removal in Stormwater: Treatment Performance and Toxicity Evaluation.
6. Al Mamun, M. R., **Kahkeci, J.**, Bunda-Fajunio, M. G., Gamal El-Din, Y., Hussain, N. A. S., Stafford, J. L., and Gamal El-Din, M. (2026). Biochar-supported FeC_2O_4 Photocatalysts for the Treatment of Stormwater Pollutants: Mechanistic Insights, and Toxicity Evaluation.

Conference Presentations

1. **Kahkeci, J.**, Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (2026, July). Tuning softwood biochar pyrolysis temperature to enhance solar-driven Bi_2WO_6 photocatalysis for organic micropollutant removal in river water. *5th International Water Association (IWA) International Conference on Disinfection and Disinfection By-Products*, Shanghai, China.
2. Gamal El-Din, M., Usman, M., Feng, Q., Bekele, A., **Kahkeci, J.**, Li, Y., Fan, Y., Arslan, M., Jaffe, B., Jessup, R. W., and Bremer, E. (2026, July). Biochar and Torrefied Biomass Soil Amendments: Influence on Nutrient Dynamics in Clayey and Sandy Soils. *1st International Conference on Land Degradation and Restoration*, Valencia, Spain.

3. Aoudi, B., **Kahkeci, J.**, Boluk, Y., & Gamal El-Din, M. (2024, August). Advanced techniques for wastewater treatment: Enhancing photocatalytic efficiency of graphitic carbon nitride. International Water Association (IWA) World Water Congress & Exhibition 2024, Canada.
4. Aoudi, B., **Kahkeci, J.**, Boluk, Y., & Gamal El-Din, M. (2024, June). Sustainable water treatment: Photocatalytic degradation of 1,3-diphenylguanidine (DPG) using graphitic carbon nitride. *Canadian Society for Civil Engineering (CSCE) Annual Conference*, Niagara Falls, ON.
5. **Kahkeci, J.**, Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (2024, June). Enhancing Solar Photocatalysis of Emerging Contaminants in Wastewater using Biochar Supported Bismuth Tungstate: Effects of Biochar Supports and Synthesis Parameters. *Canadian Society for Civil Engineering (CSCE) Annual Conference*, Niagara Falls, ON.
6. **Kahkeci, J.** and Sarathy, S. M. (2021, June). Green Hydrogen from Saline Water. *King Abdullah University of Science and Technology Research Conference: Near-Zero Carbon Combustion Technology*, Thuwal, Saudi Arabia.
7. **Kahkeci, J.** and Culfaz-Emecen, P. Z. (2018, May). Fabrication of Cellulose Membranes for Organic Solvent Nanofiltration via Alkaline Hydrolysis of Cellulose Acetate. *Gazi University Chemical Engineering Conference*, Ankara, Turkey.

Doctoral Dissertation and Master's Thesis

1. **Kahkeci, J.** (2025). *Biochar-Supported Bismuth Tungstate Photocatalysts for the Removal of Emerging Contaminants from Wastewater* [Doctoral dissertation, University of Alberta].
2. **Maskoun, J.** (2020). *Optical microfluidic waveguides and solution lasers of colloidal semiconductor quantum wells* [Master's thesis, Bilkent University].